

SCHOOL OF ELECTRONICS ENGINEERING (SENSE) · DATA SCIENCE

PROJECT KINETICA

Vision-Driven Dynamic Traffic Actuation & Emergency Preemption

A cyber-physical system unifying computer vision, stochastic traffic modeling, priority data structures, and predictive analytics to replace fixed-timer intersections.

TECHNICAL PROJECT PROPOSAL — VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI

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This document presents the complete technical rationale, mathematical formulation, and system architecture of Project Kinetica for faculty evaluation and guidance.

0 Abstract

Urban intersections governed by fixed-timer traffic signals allocate right-of-way using static, pre-programmed schedules that do not respond to actual traffic conditions. This results in avoidable congestion, wasted “empty green” time, and delayed response for emergency vehicles. **Project Kinetica** proposes a closed-loop, cyber-physical alternative that continuously observes intersection state through computer vision, models arrival and departure behavior using stochastic traffic-flow theory, and actuates signal phases through a priority-driven data structure capable of city-wide emergency corridor preemption. A predictive analytics layer validates the system's performance against classical fixed-timer baselines using regression-based forecasting and formal hypothesis testing. The result is an intersection controller that is reactive, mathematically grounded, and provably superior to static scheduling under variable traffic load.

1 Introduction & Problem Statement

Conventional traffic signal controllers operate on fixed-cycle timers calibrated from historical or average traffic counts. These schedules are blind to real-time conditions: a lane with negligible traffic may receive a full green phase, while an adjacent lane with a rapidly growing queue waits idly. Two consequences follow directly from this rigidity:

- **Inefficient throughput** — green time is allocated by schedule rather than by demand, producing both queue overflow on busy approaches and wasted capacity on light ones.
- **Unsafe delay for priority vehicles** — ambulances, police vehicles, and school transport receive no special treatment and must queue like ordinary traffic, directly impacting emergency response time and child safety.

Project Kinetica addresses both failure modes by replacing the static schedule with a perception-driven, statistically modeled, and priority-aware actuation engine. The system is organized as four cooperating modules — Vision, Actuation, Preemption, and Predictive Validation — each contributing a distinct computational discipline to a single feedback loop, illustrated below.

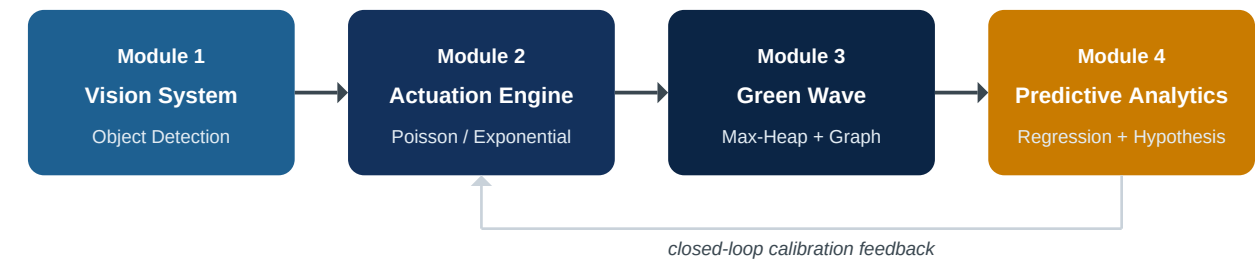


Figure 1. Closed-loop architecture of the four Kinetica modules.

2 Design Objectives

Objective	Success Criterion
Demand-responsive green time	Green phase duration scales with measured queue length, not a fixed clock.
Zero-latency emergency preemption	Detected priority vehicles trigger phase change ahead of normal queue processing.

Objective	Success Criterion
Corridor-level green wave	Consecutive downstream intersections pre-clear along the emergency vehicle's route.
Statistically validated improvement	Hypothesis testing confirms significant wait-time reduction vs. fixed-timer baseline.

3 Module 1 — The Vision System

The Vision System is Kinetica's sole sensing layer, converting raw camera feeds into structured, per-lane traffic state at every signal cycle.

3.1 Continuous Monitoring

A convolutional object-detection framework processes static traffic-camera feeds frame by frame, localizing every vehicle in the field of view without requiring specialized inductive loop sensors or radar hardware.

3.2 Queue & Density Estimation

Bounding boxes returned by the detector are projected onto lane geometry to compute two quantities per approach: **spatial density** (vehicles per unit road length) and **queue length** (the physical extent of stationary vehicles measured back from the stop line). These two measurements are the primary inputs to the Actuation Engine.

3.3 Vehicle Classification

Each bounding box is additionally passed through a classification head that assigns a priority class, allowing downstream modules to treat lanes differently based on their occupants rather than their volume alone.

Class	Priority Tier	Actuation Behavior
Standard vehicle	Baseline	Contributes to normal density/queue scoring only.
Two-wheeler	Baseline	Counted with adjusted spatial footprint in density calculation.
Ambulance	Critical	Triggers max-heap override and graph-based green wave.
Police	Critical	Triggers max-heap override and graph-based green wave.
School van	Elevated	Receives priority multiplier; escalates to critical near school zones.

Table 1. Vehicle classification taxonomy and downstream effect.

4 Module 2 — The Kinetica Actuation Engine

4.1 Statistical Modeling of Traffic Flow

Vehicle arrivals at an approach are modeled as a Poisson process, an assumption standard in traffic engineering for uncoordinated arrival streams. The probability of observing k arrivals in an interval of length t is:

$$P(N(t) = k) = (\lambda t)^k \cdot e^{-\lambda t} / k!$$

where λ is the mean arrival rate (vehicles per second) estimated in real time from the Vision System's frame-over-frame counts. The corresponding inter-arrival gaps follow an exponential distribution:

$$f(x) = \lambda \cdot e^{-\lambda x}, \quad x \geq 0$$

This pairing gives Kinetica a continuous, self-updating estimate of how “fluid” each approach's traffic is — dense, bursty arrival (small mean gap) versus sparse, free-flowing traffic (large mean gap) — without any

manual recalibration.

4.2 Dynamic Green Extension & Early Cutoff

When the Vision System reports a queue exceeding a calibrated density threshold, the engine extends the active green phase by the minimum time required to discharge that queue, estimated from the observed departure (saturation) flow rate. Critically, the extension is not fixed: the engine re-samples the queue on every processing cycle and terminates the green phase immediately once the queue clears — eliminating “empty green” time that fixed-timer systems cannot avoid.

5 Module 3 — Green Wave Preemption

This module governs cross-lane and cross-intersection prioritization using two classical data structures applied to a live traffic context: a max-heap for local phase selection and a directed graph for city-scale corridor coordination.

5.1 Priority Queuing via Max-Heap

Every incoming lane at an intersection is a node in a max-heap keyed by a live priority score. Under normal conditions the score grows monotonically with waiting time and queue density, guaranteeing that no lane starves indefinitely (an aging mechanism). The controller always actuates the lane at the heap root, giving $O(\log n)$ updates and $O(1)$ access to the next phase to serve.

5.2 Emergency Override

When the Vision System's classifier flags an Ambulance, Police, or School Van, that lane's score is multiplied by a large priority coefficient $\mu \gg 1$, guaranteeing heap-root placement on the next update regardless of current queue state elsewhere. This is an $O(\log n)$ operation, so override latency remains bounded even at large intersections with many approach lanes.

5.3 Graph-Theoretic Corridor Routing

The city grid is represented as a directed graph $G = (V, E)$, where each vertex $v \in V$ is a signal-controlled intersection and each edge $e \in E$ is a directed road segment weighted by expected travel time. Once a priority vehicle is detected entering the grid, Kinetica projects its most probable downstream path (via current heading and historical route patterns) and pre-issues green signals at each node along that path timed to the vehicle's estimated arrival — producing an uninterrupted “green wave” corridor rather than a series of independent local overrides.

Structure	Role	Complexity
Max-Heap	Local lane prioritization at a single intersection	Insert / update: $O(\log n)$ Peek root: $O(1)$
Directed Graph	City-wide corridor projection for an active emergency route	Shortest / likely path: $O(E + V \log V)$

Table 2. Data structures underlying the preemption module and their asymptotic cost.

Together, the heap guarantees an immediate, correct local decision, while the graph guarantees that decision is coordinated with every intersection the vehicle has not yet reached — converting isolated preemption into a genuine corridor effect.

6 Module 4 — Predictive Validation & Analytics

6.1 Predictive Delay Modeling

A regression-tree model is trained on historical and streaming visual/flow data to forecast two second-order effects that simple queue-clearing logic cannot anticipate: emerging bottlenecks on approaches not currently being served, and cross-traffic delay incurred as a side effect of an emergency preemption event. These forecasts allow the Actuation Engine to pre-emptively soften the impact of a green-wave event on the rest of the network rather than reacting to congestion after it has already formed.

6.2 Statistical Proof of Improvement

To validate the system rigorously rather than anecdotally, Kinetica formulates its own evaluation as a hypothesis test comparing mean intersection wait time under the dynamic controller against a fixed-timer baseline running on identical traffic samples:

$$H_0: \mu_{Kinetica} = \mu_{baseline} \quad \text{vs.} \quad H_1: \mu_{Kinetica} < \mu_{baseline}$$

A two-sample t -test (or Mann-Whitney U test where wait-time distributions are found to be non-normal) is applied to independent wait-time samples collected from both controllers under matched traffic load. Rejecting H_0 at a pre-registered significance level ($\alpha = 0.05$) constitutes the project's formal claim of improvement, rather than relying on aggregate averages alone.

7 Expected Outcomes

Metric	Fixed-Timer Baseline	Project Kinetica
Green time allocation	Static, pre-scheduled	Demand-driven, terminates on queue clearance
Emergency vehicle handling	None (queues normally)	Heap-based override + graph-based green wave
Bottleneck awareness	None	Regression-tree forecasting of downstream delay
Evidence of improvement	Not measured	Formal hypothesis testing (t-test / Mann-Whitney U)

8 Conclusion

Project Kinetica reframes traffic signal control as a continuous estimation-and-actuation problem rather than a fixed schedule. By combining real-time vision-based sensing, stochastic flow modeling, priority-aware data structures, and predictive statistical validation into a single closed loop, the system is designed to reduce average intersection wait time, eliminate wasted green time, and — most critically — provide provably faster, coordinated passage for ambulances, police vehicles, and school transport across an entire traffic corridor rather than a single junction.

Prepared for faculty review — Vellore Institute of Technology, Chennai. Reviewer feedback on module scope, dataset requirements, and evaluation methodology is welcomed prior to implementation planning.